ANALYTICAL APPROACH TO MARKETING DECISIONS

Employee Attrition Prediction

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Executive Summary

Due to its significant impact on operations, financial costs and overall loss of human capital, employee attrition has become a challenge for many organizations and their HR teams. This report carefully examines the key drivers of employee attrition by using data driven analysis on an HR dataset that contains variables such as demographics, job-related (satisfaction, environment), compensation and promotion.

The report begins with a problem statement and data dictionary to help map out the structure and variables of the dataset. We also conducted Exploratory Data Analysis (EDA) to help us assess the quality of the dataset by dealing with missing values and outliers. The methodology involved building and validating predictive models using Logistic Regression and Decision Tree algorithms. Multiple train-test splits and cross-validation were applied to strengthen validity and applicability. Evaluation of model performance was conducted using precision, recall, F1-score, ROC curves, and Precision-Recall curves across different thresholds.

We were able to identify key variables through our analysis which are mentioned in this report in greater detail and supported with evidence. Based on these findings, we have recommended policies aimed at improving the problem of employee attrition.

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Introduction

The business environment has become increasingly competitive, due to which organizations face the growing challenge of retaining their employees. Being unable to hold onto their workforce causes firms to bear high recruitment and training costs, this loss of human capital also causes them to lose their organizational knowledge. Hence, understanding and predicting employee behavior has become a priority for HR teams.

While organizations hold access to large volumes of data relating to its employees, they lack the ability to transform said data into actionable insights that would help support their decision making. This is where analytical approaches and predictive modelling can help, it can identify patterns that lead to certain behavior and uncover key drivers of attrition.

By using a dataset from Kaggle that included detailed employee information (demographic attributes, job roles, compensation, job satisfaction, work-life balance, and tenure related data), this project aims to address the problem of employee attrition. Both numerical and categorical variables were identified, allowing for the use of multi-dimensional analysis and predictive modeling.

Problem Statement

Despite the availability of HR data, most organizations lack the frameworks to proactively identify what employees are at risk of leaving. By utilizing the said data, this project tackles the issue of employee attrition, and with the help of Logistic Regression and Decision Tree models, significant key drivers of attrition have been identified. Insights derived from the analysis are intended to support HR and employee retention policies.

Data Dictionary

The dataset consisted of a categorical output variable **"Attrition."** The column indicated whether an employee left the company in the last year or not (Yes = Left the company, No = Did not leave the company). The dataset has a total of 30 input variables. The table below lists all the variables that were used in the analysis.

Variable Name	Description	Identifier
Age	Age of the employee	Numeric
Attrition	Whether the employee left in the previous year: Yes / No	Binary (Y variable)

BusinessTravel	Frequency of business travel	Categorical
Department	Department in the company	Categorical
DistanceFromHome	Distance from home (kms)	Numeric
Education	Education level: 1 Below College, 2 College, 3 Bachelor, 4 Master, 5 Doctor	Categorical
EducationField	Field of education	Categorical
EmployeeID	Unique employee ID number	Identifier
Gender	Gender of employee	Categorical
JobLevel	Job level in the company: 1 to 5	Categorical
JobRole	Job role in the company	Categorical
MaritalStatus	Are they married, single, or divorced	Categorical
MonthlyIncome	Monthly income	Numeric
NumCompaniesWorked	Number of companies worked at	Numeric
PercentSalaryHike	Salary hike percentage	Numeric
StandardHours	Standard working hours	Constant
StockOptionLevel	Stock option level: 0, 1, 2, 3	Categorical
TotalWorkingYears	Total working experience (years)	Numeric
TrainingTimesLastYear	Number of trainings attended last year	Numeric
YearsAtCompany	Years spent at the company	Numeric
YearsSinceLastPromotion	Years since last promotion	Numeric
YearsWithCurrManager	Years with current manager	Numeric
EnvironmentSatisfaction	Work environment satisfaction: 1 Low, 2 Medium, 3 High, 4 Very High	Categorical
JobSatisfaction	Job satisfaction level: 1 Low, 2 Medium, 3 High, 4 Very High	Categorical
WorkLifeBalance	Work-life balance level: 1 Bad, 2 Good, 3 Better, 4 Best	Categorical

JobInvolvement	Job involvement level: 1 Low, 2 Medium, 3 High, 4 Very High	Categorical
PerformanceRating	Performance rating: 1 Low, 2 Good, 3 Excellent, 4 Outstanding	Categorical
No. of Overtimes	How many times have they worked overtime	Numeric
Overtime	Whether the employee worked overtime: Yes / No	Binary
Total Hours Worked	Total hours worked by employee	Numeric
No. of Leaves	Number of leaves taken by the employee	Numeric

Table 1: Variables Description

Exploratory Data Analysis (EDA)

The HR Attrition dataset initially contained **4,410 rows** and **29 columns**. Then all the EDA, including data cleaning, handling NA values, and creating new variables, were done in **Excel**.

Outlier Detection

In our dataset, there weren't many numerical columns with high values. The columns **MonthlyIncome** and **DistanceFromHome** were the only columns on which outlier detection was performed. We applied the IQR to both columns and then calculated their upper and lower bounds to see how many values lie outside the bounds. For the column **DistanceFromHome**, 0 values were outside the bounds. While for **MonthlyIncome**, **340 values** were outside the bounds, which represented around 8% of the data. But upon further analysis and consultation with Sir Hassan, the salary of these outliers belonged to either high level workers or every experienced person in their field. Therefore, these were not outliers that should be discarded; hence, they were used for further analysis.

Handling Missing and NA Values

The dataset did not have any missing values but had NA values in 5 of the columns. Three of the columns were **EnvironmentSatisfaction, JobSatisfaction,** and **WorkLifeBalance.** As all of these columns are closely linked to the department an employee works in, their job role, and job level, we grouped the data based on these factors. Then, each NA value was replaced by the mode of the respective column of its own group (Department, Job Role, Job Level). Also, mode imputation was done rather than median or mean because these columns were ordinal categorical columns. The other column with NA values was **TotalWorkingYears,** for which we just used the overall median of the column.

Lastly, it was **NumCompaniesWorked**. It had 2 issues one was the NA values, and the second was that 0 companies worked, which is not possible as he/she must have worked in at least one company. So, for the NA values, we did the same as TotalWorkingYears, which was the overall median of the column. For the 0 values, we first checked if the **total working years and years at the company were the same**, so this means they had only worked for one company, but if they are not the same, then we assumed that the person changed companies each year. Therefore, the NumCompaniesWorked for those rows was calculated by the formula: **1 + (TotalWorkingYears - YearsAtCompany)**.

New Variables

Two columns, namely **Over18** and **EmployeeCount,** were removed from the original dataset as they had only one value and were not helpful variables for data analysis. But four new variables were created to further help make a good model for predicting attrition. These columns are **Overtime, No. of Overtimes, Total Hours Worked, and No. of Leaves**. Overtime was calculated by comparing the standard 8 working hours with the total hours an employee worked each day, as shown in the **Working Hrs sheet** in the Excel file.

Methodology

After performing the Exploratory Data Analysis (EDA), the dataset was randomized and then split into **80% training and 20% testing datasets** to make the model more robust. We implemented two modeling techniques in R, Logistic regression and Decision Trees, and the code was optimized using AI.

For Logistic Regression, first we used all the 29 variables, and then a **stepwise regression** was done on the initial model to retain only those variables that were significant for our model. Similarly, for the Decision Trees model, an initial model was made using the relevant R package, and then it was further **hyper tuned to avoid overfitting** and improve the generalizability.

After making the main models for both techniques, we validated the models using a **cross-validation** approach. Here, we divided the data into four testing sets, and then both models were rebuilt using the new testing and training sets and the performance for each fold was evaluated. To evaluate the performance for each fold, we used performance metrics like **Precision, Recall, and F1 score** on various thresholds to determine the optimal cutoffs. **ROC and PR curves** were also visualized for each fold. Ultimately, significant variables were identified that contributed the most to employee attrition across both models and were also justified through secondary research.

Model Performance

Our models are designed to predict the likelihood of an employee leaving the organization. To test the behavior of the models, probability thresholds were changed between 0.00 and 1.00, and the variation in precision, recall, and F1-score of both Logistic Regression and Decision Tree models was observed.

Logistic Regression:

In the case of the logistic regression model, an increase in the classification threshold led to an increase in the overall performance initially. With higher threshold intensity, greater accuracy and lower recall were observed, indicating the trade-off between false positive reduction and a higher percentage of identifying more employees who eventually leave.

On the test set, the F1-score improved steadily as the threshold increased:

- At a threshold of 0.10, the F1 score was 22.69%.
- At 0.20, it increased to 26.29%.
- The maximum **F1-score of 28.56%** was achieved at a **threshold of 0.28**.

Beyond this point, further increases in the threshold resulted in a decline in the F1-score.

Recall decreased substantially from 87.74% at a threshold of 0.10 to 64.42% at 0.28, showing that the model became more selective and missed more actual attrition cases at higher thresholds. A similar tendency was observed on the test set, where the F1-score peaked at 29.97 at a threshold of 0.29, and the recall was 67.42. This shows that the best predictive capability of the model is around a 0.28-0.29 threshold.

	ROC Curve	PR Curve
Train	0.8409	0.5670
Test	0.8452	0.5685

Table 2: AUC of ROC & PR Curve - Logistic Regression

The model shows strong discrimination with **ROC-AUC of 0.8409 (train)** and **0.8452 (test)**, indicating good generalization. **PR-AUC scores of 0.5670 (train)** and **0.5685 (test)** show moderate effectiveness in identifying true attrition cases.

Decision Tree:

The same range of thresholds was used to evaluate the decision tree model. The decision tree, however, was not very sensitive to changes in threshold, as was the case with logistic regression.

The model showed the same scores over the range of thresholds between **0.24 and 0.30**, both in the training and test sets. The performance was steady in this range, having a 52.95 precision, 47.70 recall, and 25.09 F1-score of the training set. The model had a precision of 48.41 and a recall of 41.50 with an F1-score of 22.34 on the test set.

	ROC Curve	PR Curve
Train	0.7282	0.4614
Test	0.7332	0.4311

Table 3: AUC of ROC & PR Curve - Decision Tree Model

The decision tree showed moderate discrimination with **ROC-AUC of 0.7282 (train) and 0.7332 (test)**, indicating reasonable generalization. Lower **PR-AUC scores (0.4613 train, 0.4311 test)** reflect limited ability to identify true attrition cases under class imbalance, consistent with its lower recall and F1-scores.

Interpretation and Conclusion

Logistic regression outperforms the decision tree in predicting employee attrition, offering a clear optimal threshold (0.28–0.29) that balances precision and recall and maximizes the F1-score. The decision tree shows similar performance across 0.24–0.30, with lower recall and F1, limiting its usefulness. Thus, logistic regression provides better prediction, threshold clarity, and practical value for retention decisions.

The model shows strong discrimination with ROC-AUC of 0.8409 (train) and 0.8452 (test), outperforming the decision tree (0.7282 train, 0.7332 test) and indicating better generalization. Its PR-AUC scores of 0.5670 (train) and 0.5685 (test) also exceed the decision tree (0.4613 train, 0.4311 test), reflecting improved ability to identify true attrition cases under class imbalance. Overall, this model demonstrates superior predictive performance and reliability for attrition prediction.

Significant Variables

In this project, employee attrition was modeled using both Logistic Regression and Decision Tree algorithms. After model estimation and validation, several variables consistently emerged as statistically and practically significant predictors of attrition. Evidence for their importance came from multiple sources: feature-importance scores in the Decision Tree, size and sign of coefficients in Logistic Regression, as well as observed relationships during EDA, such as group differences and correlation patterns. These variables capture dimensions of compensation, work pressure, satisfaction, career progress, and demographic background. For each of the significant variables discussed below, we interpret why it matters, how it influences the probability of attrition, and how organizations may respond from an HR policy perspective.

Years Since Last Promotion

Years Since Last Promotion emerged as one of the strongest predictors of attrition in both models. It showed a high coefficient magnitude in Logistic Regression and substantial feature importance in the Decision Tree, indicating that the longer an employee goes without promotion, the more likely they are to leave (John P. Hausknecht, 2009). This supports a straightforward behavioral mechanism: long periods without career progression signal stagnation, under-recognition, or blocked career paths. Employees with long promotion lags frequently express frustration, reduced engagement or lower levels of organizational commitment. In contrast, newly promoted workers exhibit a much lower risk of attrition. The discovery from the model supports the orthodox HR belief regarding retention, in that career prospects are at the heart of keeping talent suggesting clear promotion policies, career path communication and structured development plans.

No. of Overtimes / Total Hours Worked

Overtime frequency and total hours worked also ranked highly in the Decision Tree model, reflecting the strong association between workload pressure and attrition. Employees working excessive overtime hours exhibited significantly higher predicted probabilities of leaving. Long working hours relate directly to work-life imbalance, burnout, and emotional exhaustion. In HR analytics, overtime is widely used as a proxy for job strain. In your dataset, employees with consistently high overtime records clustered in high-attrition leaf nodes in the decision tree, providing model-based verification that chronic workload stress is a major driver of turnover. This suggests that managing workload distribution, enforcing reasonable overtime limits, and strengthening support systems are essential retention strategies (Christina Maslach, 2016).

Environment Satisfaction

Environment Satisfaction showed strong significance, particularly in Logistic Regression, where it had one of the largest coefficient magnitudes. Employees rating their environment poorly; uncomfortable workspace, poor resources, unsupportive climate, or weak culture—had far higher attrition probabilities than those who rated it highly. This is consistent with the EDA finding that attrition rates were highest among employees who selected “Low” or “Medium” satisfaction levels. The work environment operates through psychological pathways such as perceived respect, safety, belonging, and morale. The models confirm the intuitive but powerful result: employees do not leave only because of pay, many leave because of how work feels daily. Improving supervisory support, ergonomics, communication climate, and recognition systems becomes highly relevant here (Holtom, Mitchell, Lee, & Eberly, 2008).

Job Satisfaction

Job Satisfaction also appeared as a highly influential variable, particularly in the logistic regression model, where it showed a strong negative association with attrition. Employees who reported

“Low” job satisfaction were much more likely to have attrited compared to those reporting “High” or “Very High” satisfaction. Satisfaction subsumes multiple factors, including meaningful work, rewards, growth, and alignment with personal goals. Classic turnover theory suggests that declining job satisfaction acts as an intermediate psychological state that precedes turnover intentions and eventual withdrawal behavior, rather than turnover being an impulsive decision. This finding aligns with established empirical evidence that dissatisfaction initiates a cognitive evaluation process leading employees to consider and ultimately enact exit decisions. In practice, this emphasizes the importance of engagement surveys, enrichment of job roles, supervisor feedback quality, and alignment between employee skillsets and roles (Mobley, 1977).

Total Working Years

Total Working Years emerged as the most important variable in the Decision Tree model, indicating a strong influence on attrition prediction. Interestingly, the relationship is nonlinear. Employees with very low total experience show higher attrition, likely due to exploratory early-career behavior and greater external job mobility, a pattern well documented in career-stage and turnover research (Super, 1980). Very experienced employees also displayed elevated attrition in some branches, possibly related to career plateauing or retirement transitions. Mid-career employees demonstrated the lowest attrition rates. The model, therefore, highlights that career stage matters, and one-size-fits-all retention strategies are ineffective. Tailored initiatives should recognize early-career mobility and late-career exit dynamics.

Monthly Income

Monthly Income showed substantive importance in the decision tree model and clear directional effects in logistic regression. Lower-income employees demonstrated higher probabilities of attrition compared to higher-paid counterparts, reinforcing long-established evidence that compensation significantly influences turnover decisions (Trevor, 1997). This provides quantitative verification of compensation’s role as a retention lever. For many employees, especially those early in their careers, salary is directly tied to perceived fairness, market value, and financial security. The model supports a practical conclusion: uncompetitive or compressed pay bands elevate attrition risk, particularly in roles where external market opportunities are abundant. Structured pay benchmarking and performance-related compensation policies, therefore, align closely with retention objectives.

Marital Status

Marital Status appeared among the more significant variables across both models. Single employees showed higher probabilities of attrition, whereas married employees tended to stay longer, a pattern consistently observed in turnover and mobility research (Becker, 1964). This fact reflects differences in mobility, financial commitments, and stability preferences. Single employees usually are more willing to relocate, change their jobs, or pursue rapid advancement opportunities, while married employees prefer stability and predictability. The result underscores the fact that demographic context-while never used for discriminatory HR policy-can help organizations make sense of heterogeneity in turnover risk and develop targeted approaches to better engage employees.

Age

Being younger appeared to be a moderately significant factor in both models. Young employees were found to be at a distinctly greater risk of exit, and a person's probability of exiting decreased with age. This further supports the life-cycle turnover model, since younger employees are generally more mobile and prone to searching for opportunities and better offers, while older employees are apt to be more strongly attached to an organization, have more vested interests, and have benefited more from a retirement program. Age-related splits of the decision tree helped to distinguish between exits and non-existence and ascertained that age does have a predictive value.

Job Role

Job Role demonstrated measurable importance in the decision tree model. Specific roles, particularly those with sales pressure or customer-facing responsibilities, showed higher attrition rates than others. Technical and specialized roles with strong compensation and career paths showed lower attrition. This indicates that turnover risk is not uniform across an organization and is structurally embedded in role design, stress level, and reward systems. This supports role-targeted interventions such as mentoring for high-burnout roles and career pathway development for technical staff.

Years with Current Manager

Years with Current Manager appeared as a meaningful variable in logistic regression. Shorter managerial tenure was associated with higher attrition probability, suggesting that stability in supervisory relationships contributes to employee attachment, a relationship strongly supported in leader–member exchange (LMX) theory (Graen, 1995). New or frequently changing managers can disrupt trust, feedback continuity, and psychological safety. Conversely, long-term manager–employee relationships foster familiarity, support, and higher-quality exchange relationships. This confirms the managerial role as a central retention agent, emphasizing the need for leadership development and people-management training.

No. of Companies Worked

Number of Companies Worked also demonstrated one of the highest logistic regression coefficients. Workers who had previously switched jobs were also much more likely to leave again. This pattern is indicative of behavioral momentum in moving preferences: turnover tracked turnover across years. Workers with already high external mobility remain more willing to leave.

This knowledge can be used during hiring and onboarding to inform us that companies can expect to see higher attrition risk from career “movers”, allowing them the opportunity to implement proactive growth-focused retention strategies when engaging with these people.

Policy Recommendation

Our data analysis has identified the key factors that contribute to employee attrition, showing that turnover is systemic in nature and employees are leaving due to organizational friction areas and not because of a mere coincidence. Through our analysis of variables including *Years since last Promotion, Overtime* and *Environment Satisfaction*, *etc.,* we found out that career stagnation, burnout, and environmental dissatisfaction are the main causes of attrition. The following recommendations are structured to transition the organization’s retention strategy from reactive to proactive, resolving these root causes before they lead to resignation.

The analysis identified *Years Since Last Promotion* as one of the primary predictors of attrition, which means that the employees do not take the promotion merely as a title change but as a necessary confirmation of their career path. Long gaps between promotions will indicate stagnation and cause disengagement. To overcome this, the organization needs to carry out open promotion policies and development schemes. Promoting just on a tenure basis is not enough; HR should be involved in educating employees about the career path to ensure that they are aware of the milestones of progression. We also suggest adopting the concept of the “Career Lattices” in addition to the standard ladders so that it is possible to make lateral transfers between the departments to nurture growth and acquisition of skills even when there are no relevant vertical positions within the management. By eliminating uncertainties about career advancement, the organization will mitigate the frustrations that are currently pushing talent out. (Valk, 2025)

There is also a strong association between *Overtime* and attrition, which would also require an aggressive workload management strategy. The data supports the fact that job strain is a proxy of chronic overtime, and the high-overtime workers have a much higher likelihood of leaving. To solve this, the organization ought to put reasonable overtime restrictions and adopt a Cap and Review policy. To address this, the organization should enforce reasonable overtime limits and implement a "Cap and Review" policy. This will trigger a management review if an employee consistently exceeds a set threshold of hours, serving as an early warning system for understaffing or burnout. Furthermore, *Total Hours Worked* also shows a strong correlation; we suggest formalizing flexible working procedures, including remote work opportunities or compensatory time off (so-called cool-down periods) after high-intensity project cycles. This is an indication that the organization regards its employees as an asset that must be preserved and never drained.

With *Environment Satisfaction* and *Job Satisfaction* becoming highly significant predictors, it becomes evident that the daily experience of the work is what determines the attrition, and the strategic approach involving a transition of tacit, informal management styles to explicit, formal support models is needed. To overcome this, the organization will need to create a strong sense of "Cultural Fit" and "Social Identity", since the employees are much more satisfied when their own values align with the organizational objectives and they sense a strong connection with their own group. (Imonikhe, 2024) The management must promote social cultural integration and institutionalize the norms of teamwork so that employees do not merely work in parallelism but identify with their team and position. Moreover, to increase satisfaction at an individual level, the organization should establish platforms of active engagement, which include regular attendance at work meetings and chances to offer innovative solutions to emerging issues, which have been found to be sources of satisfaction. With consistent and in-depth analysis of these cultural and social requirements instead of concentrating on output only, the organization can develop a supportive, non-judgmental atmosphere that prevents turnover intention proactively. (Bangwal & Tiwari, 2019)

Monetary benefits have been one of the most important retention tools. The analysis shows that working at lower incomes, employees are more vulnerable to attrition, especially where there are external market opportunities. To overcome this, the organization should undertake systematic pay benchmarking to guarantee that the pay bands are competitive and are not squeezed. In circumstances where increases in base salary are impossible, the company must use variable compensation and, with suggested data of past years, democratize accessibility to stock options among high-performing staff in the lower ranks. This aligns personal financial success to the growth of an organization in the long term. (Blazina, 2025)

The analysis also shows that low tenure of the manager (*Years With Current Manager*) is correlated with greater attrition, which proves that the stability of supervisory relationships is a key factor in employee attachment. A high rate of leadership transition breaks feedback and trust. In such cases, the organization must invest in developing its leadership and make sure that whenever a change in management occurs, there must be a protocol to follow. (George & Poluru, 2024) We also suggest having "Stay Interviews" at the time of change of reporting lines to make sure the new manager-employee relationship is developing properly. (Robeano, 2017)

Lastly, retention strategies should be segmented. The significance of *Age* and *Marital Status* reflects that younger, single workers are more structurally mobile and more likely to leave. To overcome this, the strategy to be taken is to have rapid progression and rotation programs that would quench their desire to have new experiences. On the other hand, the predictive variable is Number of Companies Worked, which predicts future attrition, implying that past behavior is a good predictor of future mobility. During the recruitment process, this knowledge can be used by HR when applying for Realistic Job Previews. The organization can be frank about the intensity and expectations of the role in the hiring process, by which it can match the candidates with the culture without incurring the high cost of turnover in the early stages.

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Appendices

All the relevant files can be found Here.

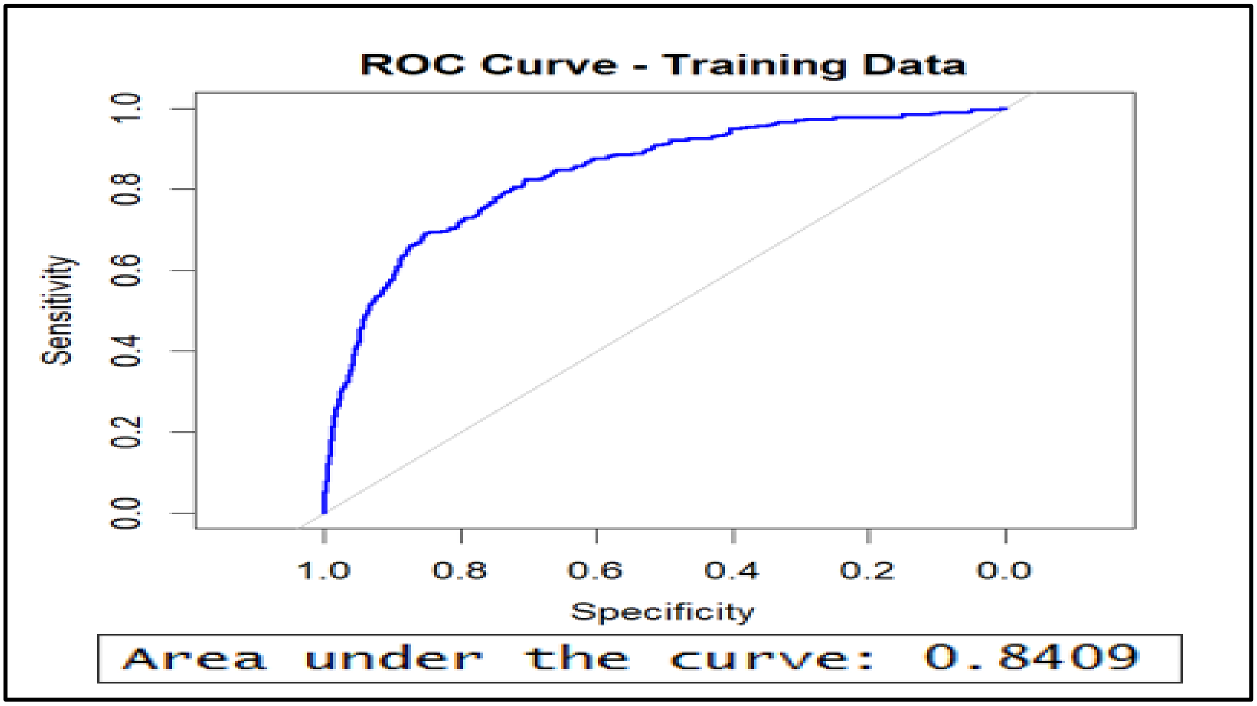

Figure 1: Train ROC Curve - Logistic

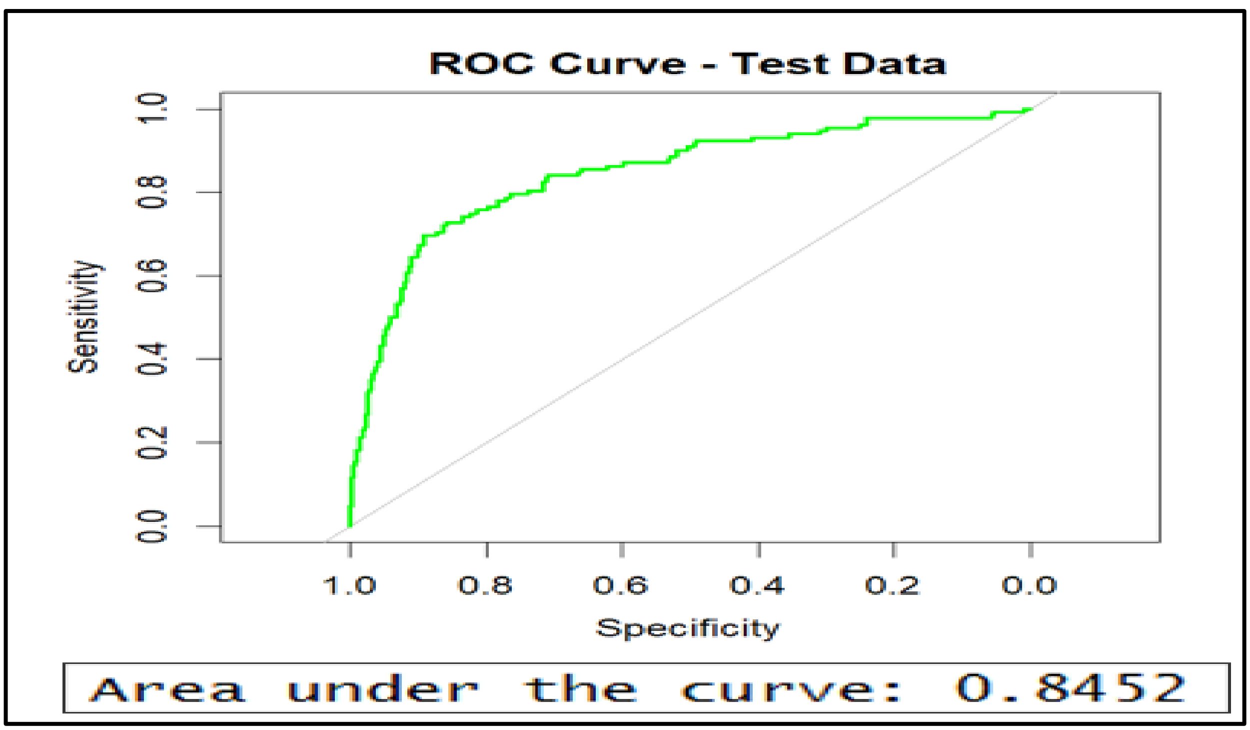

Figure 2: Test ROC Curve - Logistic

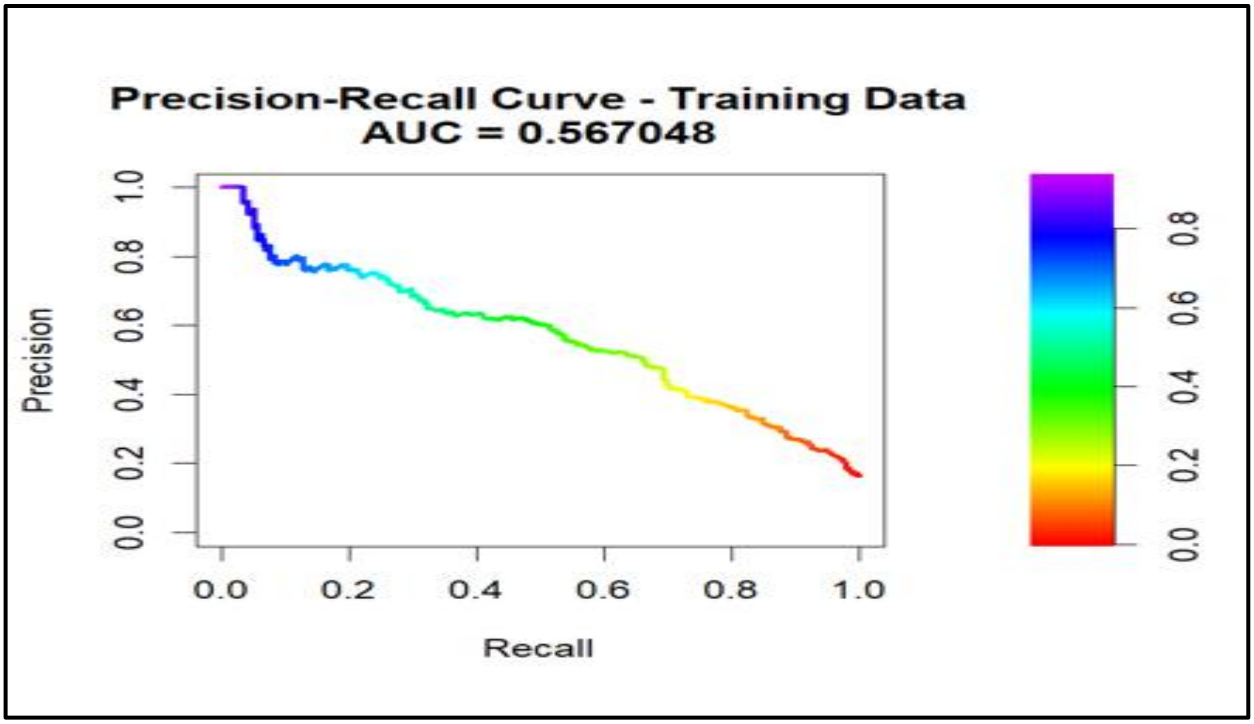

Figure 4: Train PR Curve - Logistic

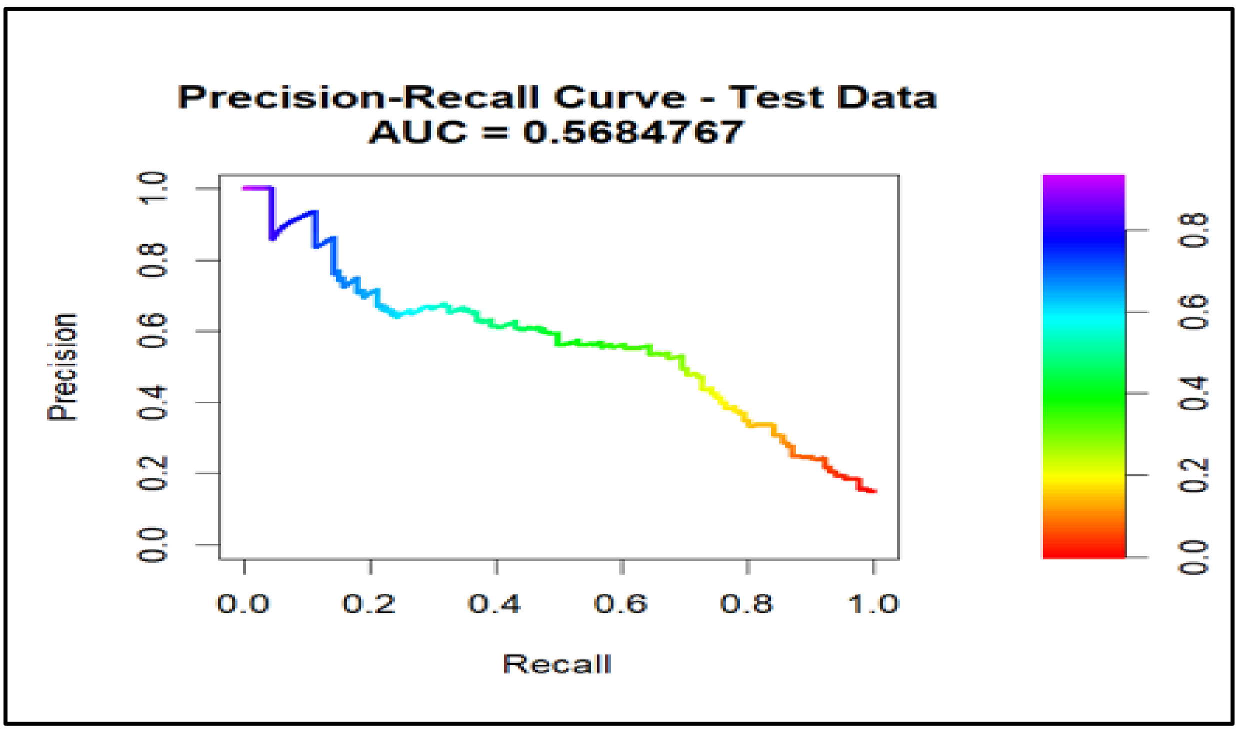

Figure 3: Test PR Curve - Logistic

```
Coefficients:
                                   Estimate Std. Error z value Pr(>|z|)    
(Intercept)                       1.4318375  0.6182254   2.316 0.020556 *  
Age                              -0.0294822  0.0082756  -3.563 0.000367 ***
BusinessTravelTravel_Frequently   1.6209386  0.2421712   6.693 2.18e-11 ***
BusinessTravelTravel_Rarely       0.7338001  0.2258411   3.249 0.001157 ** 
DepartmentResearch & Development -0.9962884  0.2252050  -4.424 9.69e-06 ***
DepartmentSales                  -1.0946661  0.2366343  -4.626 3.73e-06 ***
JobLevel2                         0.2398680  0.1245336   1.926 0.054088 .  
JobLevel3                        -0.0448291  0.1695325  -0.264 0.791450    
JobLevel4                         0.0419416  0.2191686   0.191 0.848238    
JobLevel5                        -0.4616491  0.2950078  -1.565 0.117613    
JobRoleHuman Resources           -0.3821793  0.3608875  -1.059 0.289600    
JobRoleLaboratory Technician      0.0808455  0.2222156   0.364 0.715996    
JobRoleManager                    0.0021387  0.2819446   0.008 0.993948    
JobRoleManufacturing Director    -0.6082256  0.2692303  -2.259 0.023875 *  
JobRoleResearch Director          0.6323944  0.2752656   2.297 0.021596 *  
JobRoleResearch Scientist         0.2354437  0.2162126   1.089 0.276178    
JobRoleSales Executive            0.4314230  0.2151065   2.006 0.044896 *  
JobRoleSales Representative      -0.0998807  0.2902938  -0.344 0.730795    
MaritalStatusMarried              0.1459594  0.1564144   0.933 0.350738    
MaritalStatusSingle               1.1410849  0.1567136   7.281 3.31e-13 ***
NumCompaniesWorked                0.1418645  0.0248809   5.702 1.19e-08 ***
PercentSalaryHike                 0.0220330  0.0144432   1.525 0.127136    
TotalWorkingYears                -0.0999104  0.0155996  -6.405 1.51e-10 ***
TrainingTimesLastYear            -0.1764373  0.0436515  -4.042 5.30e-05 ***
YearsAtCompany                    0.0416737  0.0220093   1.893 0.058297 .  
YearsSinceLastPromotion           0.1747084  0.0247471   7.060 1.67e-12 ***
YearsWithCurrManager             -0.1784050  0.0273545  -6.522 6.94e-11 ***
EnvironmentSatisfaction2         -0.9251244  0.1663555  -5.561 2.68e-08 ***
EnvironmentSatisfaction3         -0.9910990  0.1484244  -6.677 2.43e-11 ***
EnvironmentSatisfaction4         -1.1816770  0.1508056  -7.836 4.66e-15 ***
JobSatisfaction2                 -0.6133069  0.1642126  -3.735 0.000188 ***
JobSatisfaction3                 -0.6134981  0.1462960  -4.194 2.75e-05 ***
JobSatisfaction4                 -1.2864526  0.1582154  -8.131 4.26e-16 ***
WorkLifeBalance2                 -0.8998209  0.2177273  -4.133 3.58e-05 ***
WorkLifeBalance3                 -1.2494378  0.2023976  -6.173 6.69e-10 ***
WorkLifeBalance4                 -1.0661330  0.2549327  -4.182 2.89e-05 ***
JobInvolvement2                  -0.3874436  0.2293200  -1.690 0.091117 .  
JobInvolvement3                  -0.5376703  0.2138085  -2.515 0.011912 *  
JobInvolvement4                  -0.0991946  0.2579227  -0.385 0.700541    
No.of.Overtimes                   0.0086723  0.0007289  11.898  < 2e-16 ***
OvertimeYes                       0.4258970  0.1626749   2.618 0.008842 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 3150.0  on 3527  degrees of freedom
Residual deviance: 2325.6  on 3487  degrees of freedom
AIC: 2407.6

Number of Fisher Scoring iterations: 6
```

Figure 5: Model Summary of Stepwise Logistic Regression

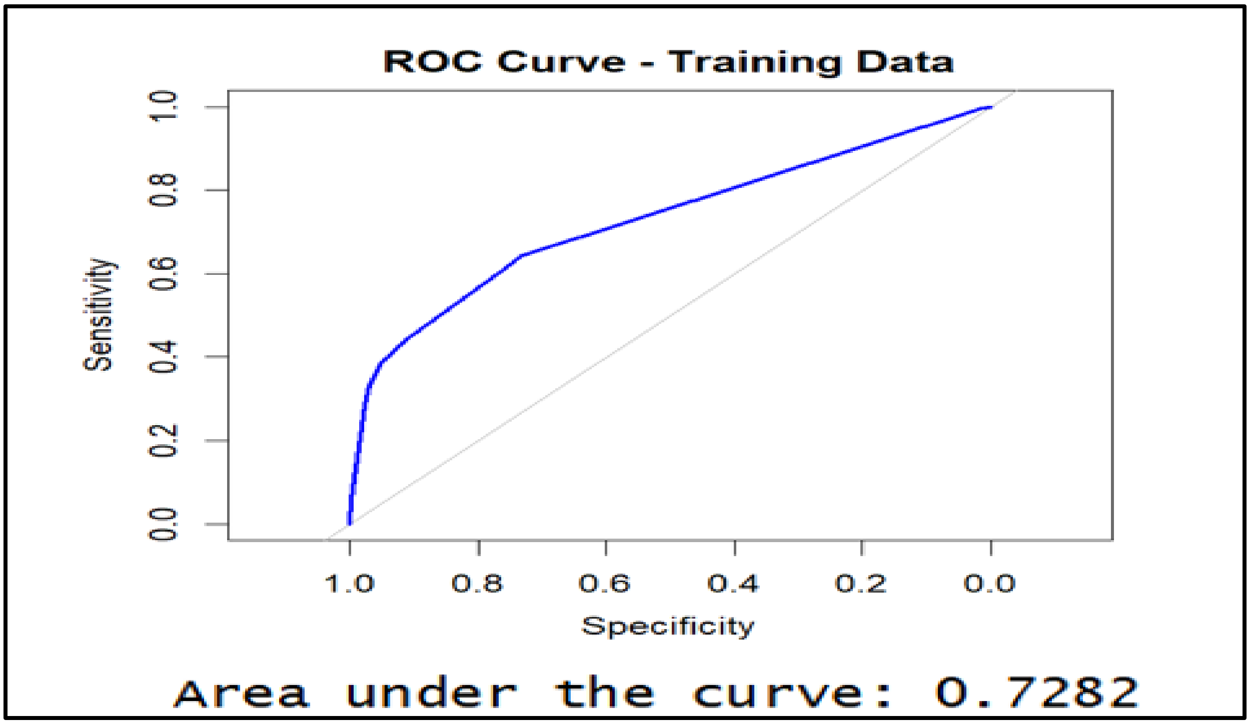

Figure 7: Train ROC Curve - Decision Tree

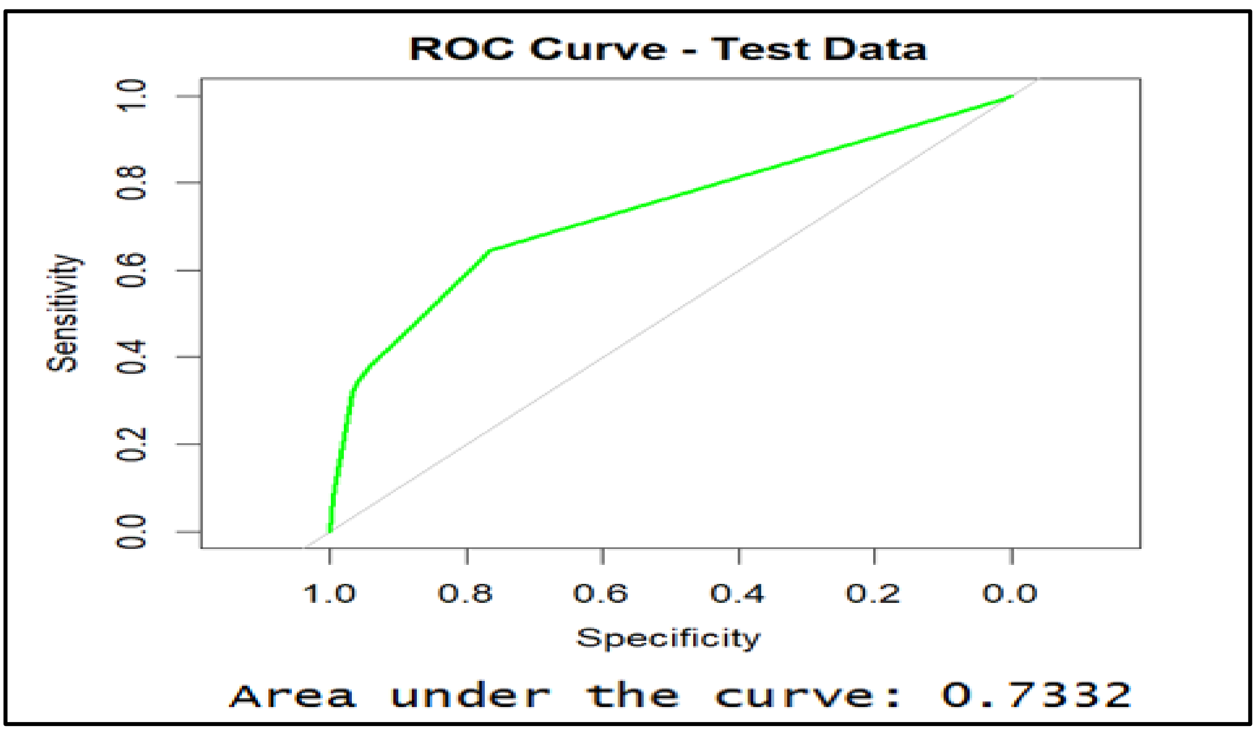

Figure 6: Test ROC Curve - Decision Tree

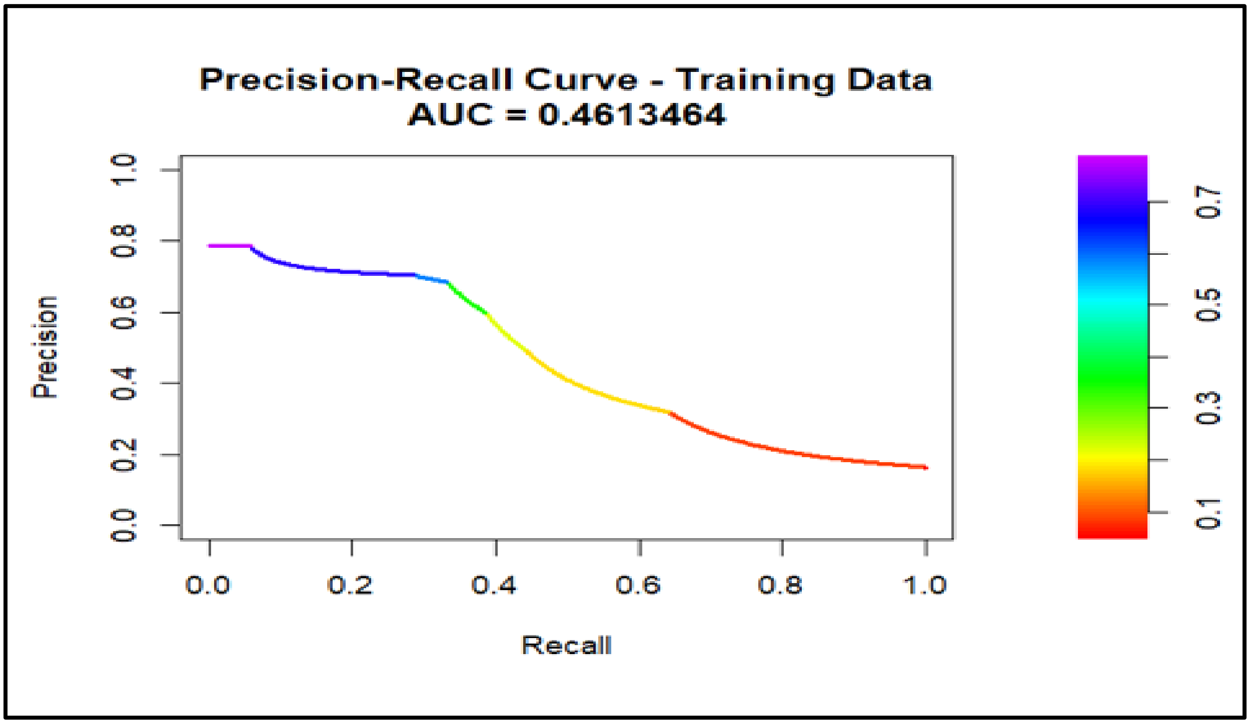

Figure 8: Train PR Curve - Decision Tree

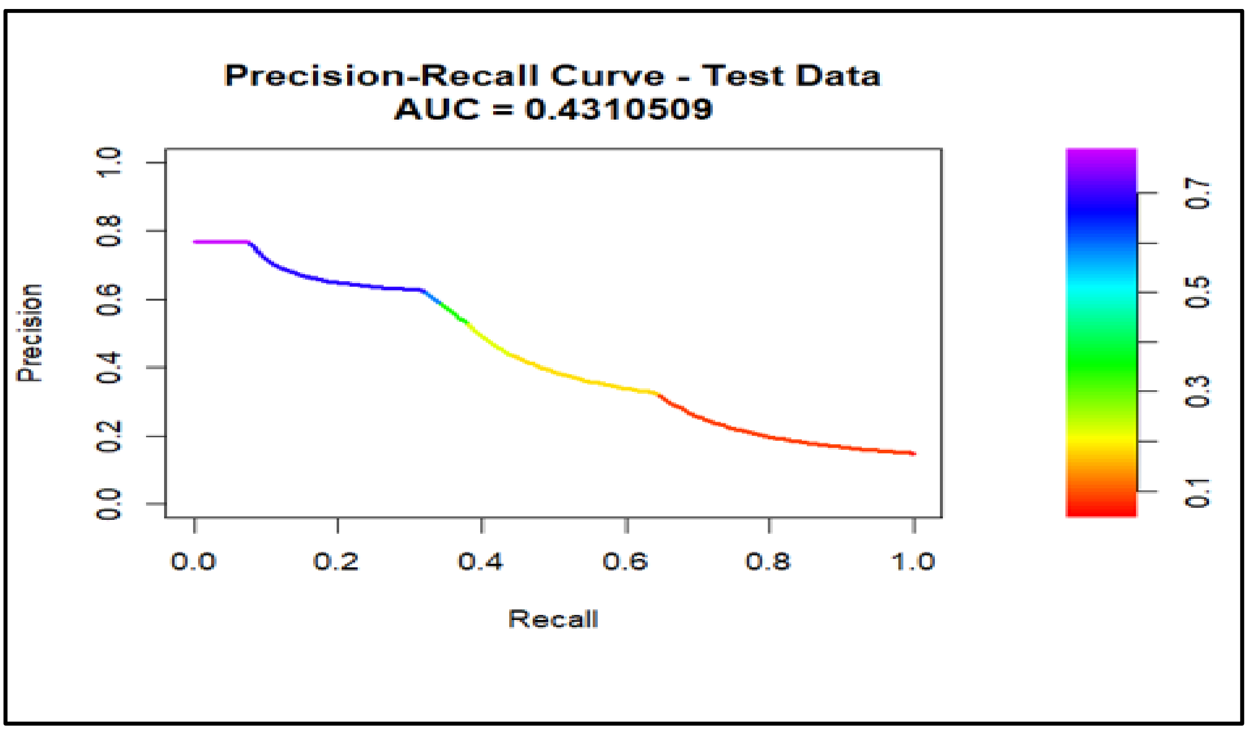

Figure 9: Test PR Curve - Decision Tree

```
> tree_model
n= 3528

node), split, n, loss, yval, (yprob)
      * denotes terminal node

 1) root 3528 579 No (0.83588435 0.16411565)
   2) No.of.Overtimes< 143.5 2550 283 No (0.88901961 0.11098039)
     4) TotalWorkingYears>=2.5 2323 205 No (0.91175204 0.08824796) *
     5) TotalWorkingYears< 2.5 227  78 No (0.65638767 0.34361233)
      10) Age>=32.5 41   2 No (0.95121951 0.04878049) *
      11) Age< 32.5 186  76 No (0.59139785 0.40860215)
        22) WorkLifeBalance=1,3 138  43 No (0.68840580 0.31159420)
          44) MonthlyIncome>=31435 92  16 No (0.82608696 0.17391304) *
          45) MonthlyIncome< 31435 46  19 Yes (0.41304348 0.58695652) *
        23) WorkLifeBalance=2,4 48  15 Yes (0.31250000 0.68750000) *
   3) No.of.Overtimes>=143.5 978 296 No (0.69734151 0.30265849)
     6) TotalWorkingYears>=8.5 559 101 No (0.81932021 0.18067979) *
     7) TotalWorkingYears< 8.5 419 195 No (0.53460621 0.46539379)
      14) MaritalStatus=Divorced,Married 275  96 No (0.65090909 0.34909091)
        28) YearsAtCompany>=3.5 139  31 No (0.77697842 0.22302158) *
        29) YearsAtCompany< 3.5 136  65 No (0.52205882 0.47794118)
          58) EnvironmentSatisfaction=3,4 94  32 No (0.65957447 0.34042553) *
          59) EnvironmentSatisfaction=1,2 42   9 Yes (0.21428571 0.78571429) *
      15) MaritalStatus=Single 144  45 Yes (0.31250000 0.68750000) *
```

Figure 10: Model Summary of Decision Tree

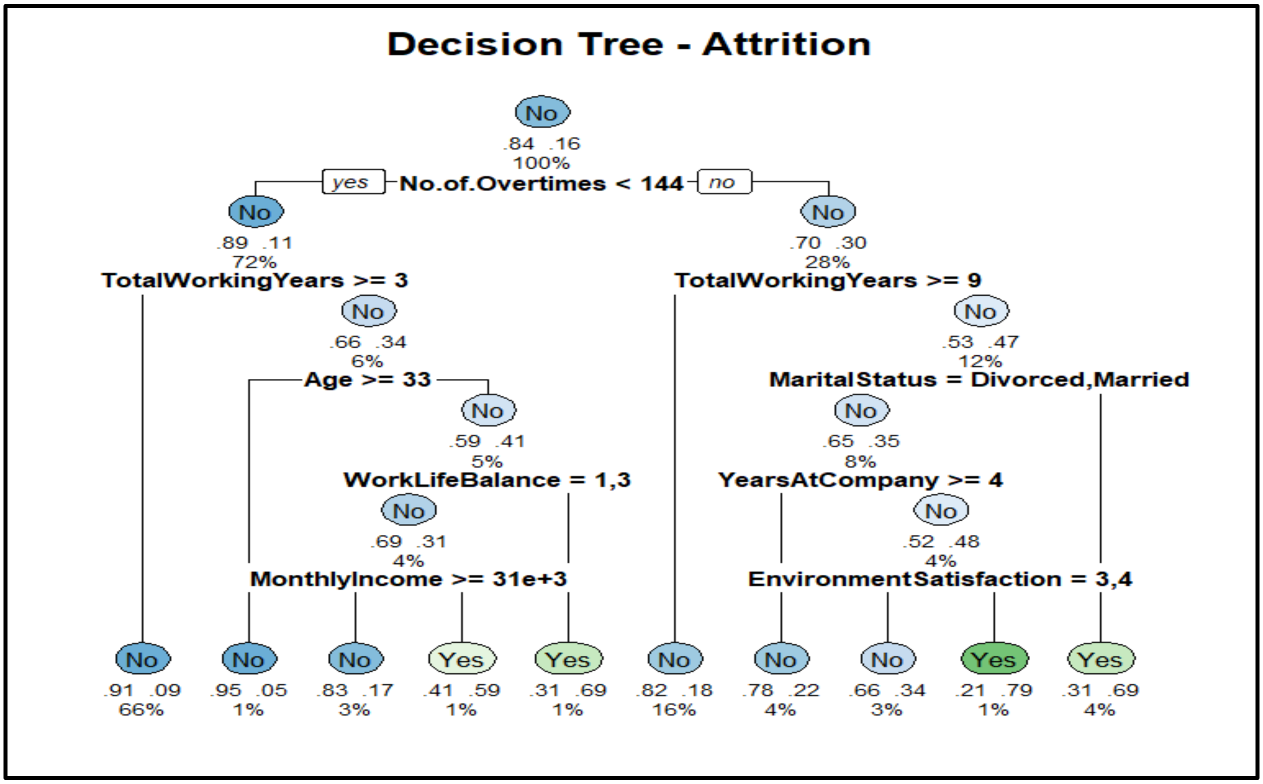

Figure 11: Decision Tree Plot

```
> tree_model$variable.importance
      TotalWorkingYears          No.of.Overtimes       Total.Hours.Worked                      Age
             71.5138320               55.2257711               50.2596288               35.7864559
         YearsAtCompany            MaritalStatus     YearsWithCurrManager            MonthlyIncome
             26.5897873               21.6468746               17.8310421               12.5281964
EnvironmentSatisfaction          WorkLifeBalance       NumCompaniesWorked             No.of.Leaves
             11.5120240               11.4350536                7.7630688                4.2489004
YearsSinceLastPromotion                  JobRole           EducationField         DistanceFromHome
              4.2487188                2.8325898                1.5211342                1.5047071
      TrainingTimesLastYear       PercentSalaryHike               Department
              1.4777310                1.0963832                0.6824197
>
```

Figure 12: Variable Importance - Decision Tree

	Threshold	True Negatives	False Positives	False Negatives	True Positives	Precision	Recall	F1-score
Train	0.28	2595	354	206	373	51.31%	64.42%	28.56%
Test	0.3	674	76	43	89	53.94%	67.42%	29.97%

Table 4: Optimal Threshold - Logistic

	Threshold	True Negatives	False Positives	False Negatives	True Positives	Precision	Recall	F1-score
Train	0.24	2725	239	295	269	52.95%	47.70%	25.09%
	0.25	2725	239	295	269	52.95%	47.70%	25.09%
	0.26	2725	239	295	269	52.95%	47.70%	25.09%
	0.27	2725	239	295	269	52.95%	47.70%	25.09%
	0.28	2725	239	295	269	52.95%	47.70%	25.09%
	0.29	2725	239	295	269	52.95%	47.70%	25.09%
	0.3	2725	239	295	269	52.95%	47.70%	25.09%
Test	0.24	670	65	86	61	48.41%	41.50%	22.34%
	0.25	670	65	86	61	48.41%	41.50%	22.34%
	0.26	670	65	86	61	48.41%	41.50%	22.34%
	0.27	670	65	86	61	48.41%	41.50%	22.34%
	0.28	670	65	86	61	48.41%	41.50%	22.34%
	0.29	670	65	86	61	48.41%	41.50%	22.34%
	0.3	670	65	86	61	48.41%	41.50%	22.34%

Table 5: Optimal Threshold - Decision Tree

Contribution Sheet

Name	ERP	Contribution
Krish Kukreja	26556	• Report Preparation and compilation. • EDA • Logistic Regression and Decision Trees
Saleem Abbas	25893	• EC 1 – Question 3 • Report Preparation • Splitting of datasets into folds in R
Fahad Ahmad	26357	• EC 1 – Question 2 • Report Preparation • Cross Validation in R
Carina Dsouza	26540	• Report Preparation • EDA • Report Formatting
Aimen Irfan	26158	• EC 1 – Question 1 • Report Preparation

Table 6: Contribution Sheet